Partial result

Separable Banach spaces have smooth non-complete norms

Status. **partial_result_likely_valid.**

Source question. In Daniel Azagra, Tadeusz Dobrowolski, and Miguel Garcia-Bravo, *Smooth approximations without critical points of continuous mappings between Banach spaces, and diffeomorphic extractions of sets*, arXiv:1811.07587, page 13, just before Lemma 2.10, the authors state that it is not known whether every infinite-dimensional Banach space with a C^1 equivalent norm possesses a C^1 smooth non-complete norm. They add that for C^k, $k \geq 2$, the answer is positive.

Result. The separable case has a simple positive answer, even without assuming the existence of a smooth equivalent norm.

Theorem 1. *Let X be an infinite-dimensional separable Banach space. Then X admits a continuous norm $|\cdot|$ such that:*

1. *$|\cdot|$ is C^∞ smooth on $X \setminus \{0\}$;*
2. *the normed space $(X, |\cdot|)$ is not complete.*

Idea of the proof. Separable spaces have countably many norm-one functionals which separate points. Weighting them by a square-summable scalar sequence gives a compact injective operator into Hilbert space. Pulling back the Hilbert norm gives smoothness. Compactness prevents this pulled-back norm from being complete: if it were complete, the compact injective operator would be bounded below, which is impossible in infinite dimension.

Proof. Choose a sequence (u_n) dense in the unit sphere S_X. For each n, pick $x_n^* \in S_{X^*}$ with $x_n^*(u_n) = 1$. The family (x_n^*) separates points: if $x \neq 0$, choose n with $\|u_n - x/\|x\|\| < 1/2$. Then

$$x_n^*(x/\|x\|) \geq x_n^*(u_n) - \|u_n - x/\|x\|\| > 1/2.$$

Define

$$T : X \longrightarrow \ell_2, \qquad Tx = \left(2^{-n} x_n^*(x)\right)_{n=1}^{\infty}.$$

Then T is bounded, injective, and compact. Indeed, the finite-rank truncations

$$T_N x = (2^{-n} x_n^*(x))_{n=1}^{N}$$

satisfy

$$\|T - T_N\| \leq \left(\sum_{n>N} 4^{-n}\right)^{1/2},$$

so T is a norm-limit of finite-rank operators.

Set

$$|x| = \|Tx\|_{\ell_2}.$$

Because T is injective, this is a norm on X, and it is continuous with respect to the original Banach norm. Since the Hilbert norm on ℓ_2 is C^∞ on $\ell_2 \setminus \{0\}$, and $Tx \neq 0$ whenever $x \neq 0$, the composition $x \mapsto \|Tx\|_{\ell_2}$ is C^∞ on $X \setminus \{0\}$.

It remains to show that $(X, |\cdot|)$ is not complete. Suppose otherwise. Then T is an isometry from $(X, |\cdot|)$ onto $T(X)$, so $T(X)$ is a closed subspace of ℓ_2. Hence $T : X \to T(X)$, viewed as a bounded bijection from the original Banach space X onto the Banach space $T(X)$, has bounded inverse by the open mapping theorem. Thus T is bounded below in the original norm.

But a compact operator on an infinite-dimensional Banach space cannot be bounded below. If it were, the compact set $\overline{T(B_X)}$ would pull back under T^{-1} to a compact set containing the original unit ball B_X, contradicting Riesz's lemma. Therefore $(X, |\cdot|)$ is not complete.

Scope and novelty check. This proves the source question for separable spaces, and actually gives a C^∞ norm. It does not address nonseparable spaces, where the proof's countable separating family is unavailable. The run indexes had no prior packet or attempt for arXiv:1811.07587 or smooth non-complete norms. Bounded web searches for "smooth non-complete norm", "C1 smooth non-complete norm", and variants found no direct later answer; the argument is elementary and may be folklore, so human review should treat novelty cautiously.

Review note. Check that the source's phrase about a smooth non-complete norm is intended in the standard sense used nearby: a continuous norm on the underlying Banach space, smooth away from the origin, whose induced normed topology is not complete. Under that reading, the proof is complete for the separable case.

Reference. D. Azagra, T. Dobrowolski, and M. Garcia-Bravo, *Smooth approximations without critical points of continuous mappings between Banach spaces, and diffeomorphic extractions of sets*, arXiv:1811.07587.